Methodologies of Software development life cycle

There are various software development life cycle models or methodologies defined and designed to follow the software development process. Each of these methodologies splits the development work into above mentioned software development phases with an intent of allowing better management and planning. Each methodology follows a series of steps unique to its type to ensure success in the process of software development. Basically, a software development methodology is a framework that is used to structure, plan, and control the process of developing an information system.

Following are the methodologies commonly used in Software Development Life cycle:

- Agile Software Development
- Crystal Methods
- Dynamic Systems Development Model (DSDM)
- Extreme Programming (XP)
- Feature Driven Development (FDD)
- Joint Application Development (JAD)
- Lean Development (LD)
- Rapid Application Development (RAD)
- Rational Unified Process (RUP)
- Scrum
- Spiral
- Systems Development Life Cycle (SDLC)
- Waterfall (a.k.a. Traditional)

Agile Software Development

Agile methodology focuses on iterative development, which ensures delivering a software system to the client quickly by developing multiple iterations of the entire system and adding more features with each iteration.

In traditional software development methodologies like Waterfall methodology, the client may not be able to see the project until its completion which may take several months or years. When it comes to projects which are non-agile, a huge amount of time is allocated for requirement gathering, design, development and testing before finally delivering the product or the software to the customer. Whereas, Agile projects have iterations which are delivered in shorter duration during which the requested features are developed and delivered. Agile projects can have a single or multiple iterations and may deliver the complete system at the end of the final iteration.